

Final Project Complex Adaptive Systems

In the final project, you will setup a small research project that is relevant given the course material. Your task is to write a report that describes your experiments and a codebase that generates results. We highly recommend that you write your code in JAX, but this is not mandatory. The submission deadline of the project is 10th of January, 13:30.

The final project constitutes a report including introduction, methods, results, discussion and when relevant citations. The report should be maximally four pages long (excluding citations) in the template below. The report should be supplemented with code (python file(s) and/or jupyter notebook) which generates the results of your report.

The final project will be evaluated on the following criteria (2.5 points each):

- clear introduction with research question and description of the methods
- quality of experiments and results
- reflection and discussion on the results
- quality of the implementation (clarity, efficiency, documentation)

When writing the report you may use the NeurIPS paper template: <https://www.overleaf.com/latex/templates/neurips-2023/vstgtvjwgdng>. If you prefer a different template, note that the page limit is based on the NeurIPS template, and the length of the report should not significantly differ.

When working on this project, prioritize clarity and solid execution over unnecessary complexity. A well-written report showcasing a straightforward and effective method is to be preferred over a complex, fancy method with unclear results and vague writing.

Below you may find suggestions of possible topics:

- *System identification under partial observability:*
Investigate to what extent neural dynamical models can handle partial observability. Take a domain of choice and show you can identify and interpret the system, without having access to the full observation space or the true dynamics. For example, possible approaches are to add an encoder and decoder structure or a Kalman filter to the dynamical model.
- *Model-based control*
Show how we can control a system using the model-based approach, for example by optimizing actions according to the Model Predictive Control (MPC) principle. A known dynamical model could be used for simulation.
- *Deep Reinforcement Learning with Q learning:*
Investigate to what extent the deep Q learning algorithm with a replay buffer and epsilon-greedy exploration, can be used to solve the pendulum swing-up (or similar) task.
- *Deep Reinforcement Learning via Evolution*
Investigate to what extent evolutionary methods, such as genetic algorithms, the Cross Entropy Method or Evolution Strategies, can be used to optimize a neural policy.
- *Deep Reinforcement Learning in practice:*
Investigate how well a deep policy gradient method, such as your REINFORCE algorithm, can solve a more challenging environment of choice (e.g., see the gymnasium or Brax libraries).

These are primarily intended as suggestions for projects. If you already have an idea that you would like to work on but is not directly described above, we recommend to discuss the feasibility and/or relevance of the project with the TAs.